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Arabic Handwritten Recognition Using Deep Learning: A Survey

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Abstract
In recent times, many research projects and experiments target machines that automatically recognize handwritten characters, but most of them are done in Latin. Recognizing handwritten Arabic characters is a complicated process compared to English and other languages as a nature of Arabic words. In the past few years, deep learning approaches have been increasingly used in the field of Arabic recognition. This paper aims to categorize, analyze and presents a comprehensive survey in Arabic handwritten recognition literature, focusing on state-of-the-art methods for deep learning in feature extraction. The paper focuses on offline text recognition, with a detailed discussion of the systematic analysis of the literature. Additionally, the paper is critically analyzing the current literature and identifying the problem areas and challenges faced by the previous studies. After investigating the studies, a new classification of the literature is proposed. Besides, an analysis is performed based on the findings, and several issues and challenges related to the recognition of Arabic scripts are discussed.

Keywords Arabic Handwritten Recognition (AHR) · Deep learning · Arabic script · Feature extraction · Classification · Arabization

1 Introduction

Handwritten language has been an inevitable aspect of human life from the beginning of time. No matter how digital the world becomes, handwritten language is a classic form that will always remain. With 315 million native speakers [1], Arabic is one of the top five most spoken languages globally, which means that computers also need to understand what these speakers are writing now more than ever. Even though the world is moving to a digital era, there are still some scenarios where the use of paper and pen cannot be avoided.

Character recognition technologies provide the users an automatic mechanism for recognizing the text on an image and converting the characters to their corresponding digital format. They are used in many verification applications, e.g., verifying official documents and bank cheques and helping visually impaired people to read, e.g., reading paper currency or street signs.

Handwriting varies, especially when it comes to the cursive script, where the writer's handwritten characters' size and style are different. Therefore, handwriting recognition is considered a more difficult task in computer vision [2]. It has been under study for a long time, with different researchers having many methods at their disposal. However, the problem is that the majority of these experiments are conducted on Latin script, e.g., English, as it is the most popular language in the world [3]. However, other languages, such as Arabic, Mandarin, and Russian, also need to have research done on them since many people speak them.

There are many available applications for handwriting recognition for well-studied scripts such as Latin. However, due to the Arabic script's unique characteristics, e.g., its cursive nature, diacritics and diagonal strokes, developing an Arabic character recognition system is still challenging [4].

Many Machine Learning (ML) algorithms have been successfully used for Arabic Handwritten Recognition (AHR). However, AHR technologies can be significantly improved by introducing Deep Learning (DL) architectures. Indeed, AHR models that use Convolutional Neural Networks (CNNs) have achieved excellent results in recognizing handwritten script [2, 5, 6].

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Table 1 Published works in AHCR and AHWR

Year	Papers in AHCR	Papers in AHWR	No. articles
2015	[10, 11]	–	2
2016	[12–15]	[16–18]	7
2017	[5, 19–23]	[24, 25]	8
2018	[6, 26]	[27, 28]	4
2019	[29–32]	[33–36]	8
2020	[2, 37–40]	[41–43]	8
Total	**23**	**14**	**37**

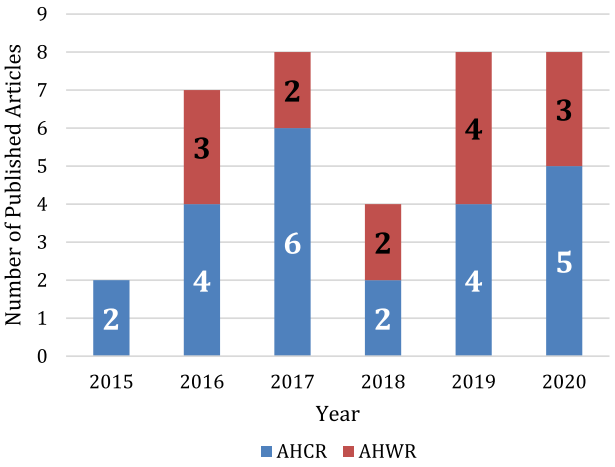


Fig. 1 Distribution of AHCR and AHWR publication (2015–2020)

Many surveys have been conducted in the field of Arabic handwritten recognition. Ahmad et al. [7] reviewed nine studies of different deep learning approaches for recognizing Arabic handwritten characters, digits, and words. The literature in this review was published between 2015 and 2017. They present a comparative study that shows the algorithm that is used, the dataset and its type, and the recognition results. Also, they conducted an experimental comparison of the performance of state-of-the-art models on different datasets. Ghadhbet et al. [8] focused on the feature extraction and classification techniques in Arabic handwritten recognition. They reviewed 21 studies: 5 studies used CNN for feature extraction, and the remaining 16 studies used classical ML for feature extraction. They also presented the datasets that are used to implement these models. Ali et al. [9] focused on the preprocessing and segmentation stages of the recognition systems.

In this survey, we review different literature (as seen in Table 1 and its baseline with the ascenders and descenders shown in Figure 1) related to deep learning approaches used in feature extraction, including a new classification for the literature. Accordingly, the contributions of this survey paper are:

- Presenting the Arabic characteristic according to characters, words, and sentence levels
- Presenting a comprehensive review of all the works reported in Arabic Handwritten recognition that use deep learning approaches
- Presenting the details of the datasets that have been used to develop the AHR deep learning models with the URL to download them
- Analyzing the related work of AHR and presenting the significant findings
- Presenting the challenges and future perspective that can enhance the AHR systems

The rest of this paper is organized as follows: section 2 describes Arabic characteristics giving a detailed description of special Arabic characteristics. Section 3 illustrated the stage of text recognition systems. Section 4 describes Arabic handwritten datasets. Section 5 presents the datasets used to develop the DL models for the Arabic recognition system in the literature. Section 6 exhaustively explains and compares the work done on Arabic handwritten recognition that employs different deep learning approaches. The state-of-the-art analysis is presented in section 7. Section 8 presented the identified challenges and open problems. Section 9 concludes the paper.

2 Arabic Characteristics

Arabic script is used in many languages, e.g., Urdu and Jawi. As opposite to many languages, Arabic is written from right to left. The Arabic characteristics will be discussed according to 3 levels: the characters levels, the words level, and the sentence and the grammatical level.

2.1 Character Level Characteristics

Arabic has 28 primary Arabic characters with no capital or small forms; however, each character is represented by 2 or 4 shapes depending on its position in the word. Therefore, the 28 Arabic letters are represented by 84 different shapes. Table 2 illustrates Arabic letters in their four forms. The first column represents the letter order in the Arabic Alphabet, the second column represents how the letter is pronounced, and the third to the sixth columns represent how the letter is written at the beginning, middle, and end of the word and the letter in the isolated form, respectively. The separating characters, e.g., ء "ا" and "ر" have only two forms: isolated and end.

- Dots are integral parts of Arabic letters. Arabic uses 16 main strokes, and dots play an important role in distinguishing the letters with the same strokes. The same stroke

Table 2 Arabic Characters in the 4 Forms: Isolated, Beginning, Middle and End

No.	Pronunciation	Isolated	Beginning	Middle	End	No.	Pronunciation	Isolated	Beginning	Middle	End
1	Alef	ا	ا	ا	ـا	15	Dhad	ض	ضـ	ـضـ	ـض
2	Baa	ب	بـ	ـبـ	ـب	16	Ta	ط	طـ	ـطـ	ـط
3	Taa	ت	تـ	ـتـ	ـت	17	Thaa	ظ	ظـ	ـظـ	ـظ
4	Thaa	ث	ثـ	ـثـ	ـث	18	Ain	ع	عـ	ـعـ	ـع
5	Jeem	ج	جـ	ـجـ	ـج	19	Ghain	غ	غـ	ـغـ	ـغ
6	Haa	ح	حـ	ـحـ	ـح	20	Faa	ف	فـ	ـفـ	ـف
7	Khaa	خ	خـ	ـخـ	ـخ	21	Qaf	ق	قـ	ـقـ	ـق
8	Dal	د	د	ـد	ـد	22	Kaf	ك	كـ	ـكـ	ـك
9	Thal	ذ	ذ	ـذ	ـذ	23	Lam	ل	لـ	ـلـ	ـل
10	Raa	ر	ر	ـر	ـر	24	Meem	م	مـ	ـمـ	ـم
11	Zai	ز	ز	ـز	ـز	25	Noot	ن	نـ	ـنـ	ـن
12	Seen	س	سـ	ـسـ	ـس	26	Ha	ه	هـ	ـهـ	ـه
13	Sheen	ش	شـ	ـشـ	ـش	27	Waw	و	و	ـو	ـو
14	Sad	ص	صـ	ـصـ	ـص	28	Yaa	ي	يـ	ـيـ	ـي

Table 3 Main Arabic Characters Strokes

Arabic main stroke	آ ب ح د ر س ص ط ع ف ك ل م ن ه و ى

Table 4 Arabic Characters That Have One, Two, or Three Dots

No. Dots	Letters									
One dot	ن	ف	غ	ظ	ض	ز	ذ	خ	ج	ب
Two dots	ي	ق	ت							
Three dots	ش	ث								

is used in different characters, but the number of dots determines the character class. The letter can have zero and up to three dots. Dots can be placed above the letter, e.g., "خ", below the letter, e.g., "ب", or inside the letter, e.g., "چ". Table 3 shows the 16 main strokes of the Arabic script, and Table 4 shows the letters with one, two, and three dots.

Hamzah "ء" is a special character that can be used as a primary character or as a subcomponent below, inside and above other primary characters [7], e.g., in Kaf "ك" and in "أ، إ ، ئ ، ؤ".

2.2 Word level characteristics

- Arabic is a cursive script in both printed and handwritten forms, which means that the letters in words are connected by a horizontal line called the baseline. The parts of characters drawn below the baseline are called "descenders," and those above the baselines are called "ascenders". Figure 1 depicts the baseline with the ascenders and descenders.
- Arabic has six separating characters "ا،ر،ز،د،ذ،و" that are not connected to their following characters. They divide the words into sub-words or Parts of Arabic Words (PAWs). Each word is represented as one or more PAWs depending on the presence of separating letters in the word. Examples of words composed of one, two, three, four, or five PAWs are presented in Table 5.

Table 5 Examples of the Words that Have One, Two, Three, Four, and Five PAWs

No. of PAWs	Example
One PAW	غ نى
Two PAWs	أحمد
Three PAWs	ذ ورة
Four PAWs	إب راهيم
Five PAWs	اوراد

- A Ligature is a special form of Arabic drawing where the second character horizontally appears before the first character. For example, when the character Lam "ل" appears before the letters Alf "ا", Meem "م", Jeem "ج", Haa "ح" or Khaa "خ" at the beginning of PAWs, a ligature is formed [38]. The Ligatures are exceptional cases and applied depending on the writing style. Table 6 shows examples of ligatures.
- "Tashkeel" or "Harkat" and "Tanween" are the diacritics used in Arabic to illustrate the letters' pronunciation and the meaning of the words [44]. There are five symbols used

Table 6 Examples of Ligatures

Ligature	Example
Lam Alef Ligature	لا
Lam Haa Ligature	لـحن
Lam-meem Ligature	لـمن
Lam – Jeem Ligature	لـجـين



Fig. 2 Vertical Overlapping (**a**) between 2 PAWs in Different Words (**b**) PAWS in one word

in Tashkeel: Fatha "َ", Dhamma "ُ", Kasra "ِ", Sokoon "ْ" and Shddah "ّ". All Tashkeel symbols are placed above the letters, except the Kasrah, which appeared below the letter. The Tashkeel can change the meaning of the words. For example, the vocabularies "عَلَم" and "عِلْم" use the same characters but are different in Tashkeel, the first means "Flag" while the second means "Knowledge." Tashkeel is not broadly used in Arabic because the reader can realize the pronunciation and meaning of the words based on context. They are usually used for novice and non-Arab readers.

- The handwritten PAWs may vertically overlap. Figure 2 shows the vertical overlapping between the characters in one word and the characters of different words.

2.3 Sentence Level Characteristics

Arabic is a rich language that uses different vocabularies to describe nouns and verbs accurately. Arabic reflects grammatical numbering (singular, dual, or plural), grammatical gender (masculine or feminine), and indefinite or definite. Also, the dual and plural nouns take more than one form based on the sentence parsing. Table 7 represents the noun "teacher" and "teachers" in Arabic. The first row represents if the grammatical numbering, the first column represents the grammatical gender in indefinite and definite forms. Table 8 presents the verb "teach" in the present, past, and imperative forms, for masculine and feminine, with respect to grammatical numbering.

Table 7 Vocabularies in Arabic of the Noun "Teacher"

	singular	dual	plural
indefinite masculine	معلم	معلمان معلمين	معلمون معلمين
indefinite feminine	معلمة	معلمتان معلمتين	معلمات
definite masculine	المعلم	المعلمان المعلمين	المعلمون المعلمين
definite feminine	المعلمة	المعلمتان المعلمتين	المعلمات

3 Stages of Text Recognition Systems

The character recognition system can be broken down into five main stages, as illustrated in Figure 3. It starts with acquiring the images that contain the text from the repository; this is the most crucial step in any machine learning project or research because using the wrong dataset renders the whole project invalid and leads to ineffective results. Machine learning and deep learning models require a lot of qualitative data. Still, the data available for experiments are often insufficient and/or not standard (in terms of size, orientation, color, format, etc.). This data needs to be cleaned and reconfigured, which leads to the next step. The preprocessing stage, where different operations are applied to the images, e.g., binarization and removing the noise, transformed the images to a more useful form. This stage involves cleaning the images to remove anomalies, transforming the images either by size or orientation, and carrying out standardization so the acquired dataset can be consistent and easily usable for the task at hand.

The paragraphs are segmented into lines in the segmentation stage, and the lines are segmented into words. After segmentation, each individual word is seen as an independent entity entered into the machine learning model. There are two types of segmentation approaches, analytics and holistic. The analytic approach or the segmentation-based approach works by segmenting the words further into separated characters. At the same time, the holistic approach is used for recognizing the whole word without segmenting it into characters.

The fourth stage is feature extraction, defined in [45] as "extracting from the raw data information, which is most relevant for classification purposes, in the sense of minimizing the within-class pattern variability while enhancing the between-class pattern variability." Because the classifier's performance highly depends on the extracted feature, it is considered one of the most important and challenging stages [46]. Feature extraction facilitates the segregation of key features in the images, which could be distinctly used to categorize various characters and words in the Arabic language. Classification is the process of identifying and recognizing the word or character by comparing its features to one of the classes [38]. This is the final stage of the pipeline; it takes the

Table 8 Vocabularies in Arabic of the Verb "Teach"

	singular	dual	plural
present masculine	يعلم	يعلمان	يعلمون
present feminine	تعلم	تعلمان	تعلمن
past masculine	علم	علما	علموا
past feminine	علمت	علمتا	علمن
Imperative masculine	علّم	علما	علموا
Imperative feminine	علّمِ	علّما	علمن

Therefore, using a dictionary to post-process the recognized word is not that efficient since the word depends on its position in the sentence



Fig. 3 Text Recognition Stages

extracted features and inputs them into the machine learning classifiers to find a pattern that could be used to categorize and classify the Arabic handwritten words.

The above approach is harder to process in the Arabic language when compared to other languages because of its extreme cursive nature. The presence of diacritics makes the process of classification correctly very hard since a single dot could change the meaning entirely.

4 Arabic Handwritten Datasets

Datasets play a significant role in building and evaluating machine learning models. Building robust handwriting recognition systems involves training them on large and diverse datasets. Also, using standard datasets is preferred since they provide an unbiased platform for evaluating and comparing the competing approaches. Offline handwritten datasets are developed by requesting volunteers to fill paper forms with digits, letters, words, sentences, or paragraphs using pen or pencil. The papers are then digitized using scanners, then the digital forms are segmented and labeled to generate the ground truth [47]. In the past two decades, a number of online and offline handwritten datasets have been developed for the Arabic script. This section presents a detailed description and statistics of various benchmark datasets that have been used in developing deep learning models for handwritten Arabic recognition systems.

Handwritten Arabic character datasets are developed to build robust and accurate recognition systems. This section presents offline datasets of isolated handwritten characters and handwritten words to develop deep learning models to recognize handwritten Arabic.

4.1 Isolated Arabic Handwritten Characters Datasets

- AlexU Isolated Alphabet (AIA9K) [48] is a dataset of handwritten Arabic characters in the isolated form developed by a research group from Alexandria University. The dataset consists of 8,737 images of Arabic letters written by 107 different writers.
- The Arabic Handwritten Characters Dataset (AHCD) [49] is one of the largest and most famous datasets in the field of offline handwritten Arabic character recognition. It was collected by El-Sawy et al. and made available online at [49]. In this dataset, 60 writers completed a form of isolated Arabic letters, and then these forms were scanned and segmented automatically to generate 16,800 images.
- Hijja [2] is the first Arabic Handwritten Character dataset aimed at advanced research to recognize children's handwriting. In the data collection phase, the children were requested to complete a 9 12 matrix with four shapes of the 28 Arabic letters, i.e., letter at the beginning of the word, middle, end, and isolated. Five hundred ninety-one forms were scanned and preprocessed to extract 47,434 images. The forms were scanned at 300 dpi resolution, while the extracted images were rescaled to be 32 32 pixels.
- Lawgali et al. [50] developed HACDB, the first dataset of handwritten Arabic characters that contains all the shapes of Arabic letters, i.e., the letters at the start, middle, and end of the word and isolated. It contains 6,600 images of 66 unique shapes written by 50 different writers. The images have only the primary stroke of the characters; i.e., the dots and diacritics are removed.
- Boufenar et al. [5] developed the OIHACDB-40 dataset by expanding their previous OIHACDB-28 dataset with 550 samples to each class.
- HMBD dataset [51] is an Arabic handwritten dataset that consists of 54,115 images of Arabic characters in different

positions: start, middle, end, and isolated; in addition to Arabic digits. The dataset files are stored in Python pickle files to enable their use with different image sizes and options of data augmentation; i.e., the image size can be 32 32, 6,4 64, or 128 128.

Table 9 summarizes the details of each dataset in terms of the year of publication, the number of writers, the statistics of the number and size of the images, and the URL of the dataset, if available.

4.2 Arabic Handwritten Words Datasets

- IFN/ENIT [52] Database is a Handwritten Arabic dataset of Tunisian town/village names that are developed by the Institute for Communications Technology (IFN) and Ecole Nationale d'Ingénieur de Tunis (ENIT). Four hundred eleven different writers participated in filling 2,265 forms of selected names from a collection of 946 Tunisian town/village names with the postcode. The forms were scanned, and the words with the ground truth data were extracted automatically, and the writers' information is also added. Then these data were verified manually. Each name may consist of more than one word with a total of 26,459 handwritten words and 115,585 PAWs (Part of Arabic Words).
- AHDB [53] is an Arabic Handwritten words dataset developed by Al-ma'adeed et al. it consists of Arabic handwritten words and free Arabic handwritten sentences and pages. They searched for the top 20 common words in Arabic. They designed a Six-page form with 96 words of the common 20 Arabic words and the literal amounts used in cheques. The word and writer's identification numbers are saved with every image.
- SUST-ARG NANES [54] is a dataset collected by (Sudan University of Science and Technology Arabic Language Technology group). The dataset is a combination of 4 subsets: Arabic numbers, Arabic characters, Arabic names, and an online Arabic dataset. The authors identified the most common male/female names and then designed 20 female names and 20 male names to develop the Arabic name subset. One thousand writers participated in creating a dataset of 40,000 handwritten word images.
- KHATT [55] is an offline Arabic handwritten dataset. One thousand writers participated in constructing the dataset where each writer had to write four paragraphs: the first is a paragraph that covers all the shapes of Arabic letters, and the writer had to write it twice; the second is a paragraph that is extracted from different forms, the third is free writing paragraph. A total of 2000 paragraphs were extracted and segmented to 6712 lines automatically then verified

Table 9 Handwritten datasets used in deep learning of Arabic handwritten character recognition

Dataset	Year	Type	Writers	Statistics	URL
AIA9K [48]	2014	Letters in isolated form	107	8737 images of Arabic letters	Available on request
Arabic Handwritten Characters Dataset AHCD [49]	2017	Letters in isolated form	60	**16,800 images** 300 dpi 32×32 Pixels	https://www.kaggle.com/mloey1/ahcd1
Hijja [2]	2020	Letters in all shapes	591	47,434 characters 300dpi Images size 32*32 pixels	https://github.com/israksu/Hijja2
HACDB [50]	2013	Arabic letter in all shapes	50		Not available
OIHACDB-40 [5]	2018	Letters in isolated form + special letters, e.g., Hamzah (ئ ،ؤ ،ء) and Taa Marbotah (ة)	–	**30,000 images**	http://www.mediafire.com/file/wks8xgfs0dmm4db/OAHCDB-40_30000_Exemples.rar/file
HMBD [51]	2020	Letters in all shapes	125	**Form of 7 pages** **Characters in 4 shapes and digits** 115 classes (characters and digits) 54,115 characters	https://www.kaggle.com/hossammbalaha/hmbd-v1-arabic-handwritten-characters-dataset/version/3

Table 10 Handwritten datasets used in deep learning of Arabic handwritten word recognition

Dataset	Year	Type	Writers	Statistics	URL
IFN/ENIT [52]	2002	Words	411	946 classes (names Tunisian town/villages) 26,400 words 210,000 characters Scanned by 300 dpi	Upon request
AHDB [53]	2004	Words	100	96 words in each form and free writing pages Scanned by 600 dpi	https://handwriting.datasetsonline.com/
SUST-ARG names [54]	2016	Characters, digits, and words	1000	40 classes (20 females name and 20 male names) Images size 28×56 pixels	Not available
KHATT [55] [56]	2012	Sentences	1000	589,924 characters 16,589 words 6712 lines 2000 paragraphs Scanned by 200, 300, and 600 dpi	http://khatt.ideas2serve.net/KHATTAgreement.php

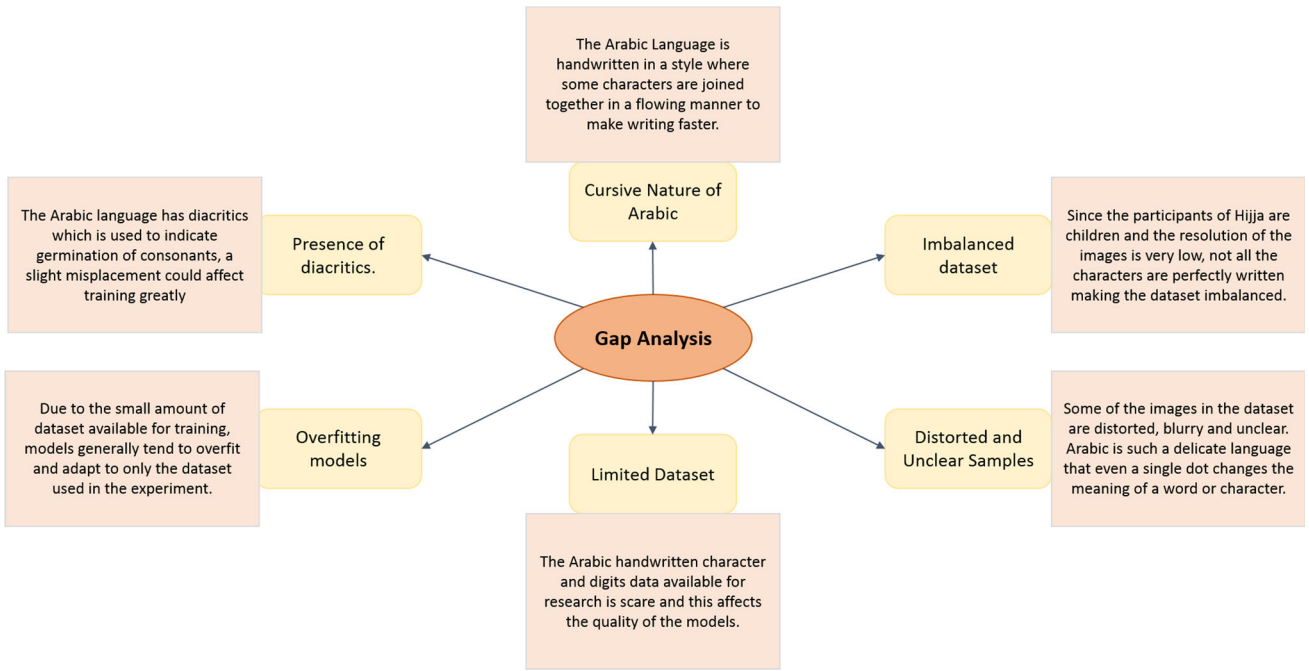


Fig. 4 Analysis of gaps encountered during the course of the research

manually. Table 10 illustrates the details and statistics of the handwritten Arabic word datasets.

In Figure 4, the gaps encountered during the course of our experiments are discussed. These gaps constitute the hindrances that will most likely cause problems for the classification models to be designed. The gaps are self-described which contains as follows:

- Cursive nature of Arabic
- Presence of diacritics
- Overfitting models
- Limited Dataset
- Distorted and unclear samples
- Imbalanced dataset

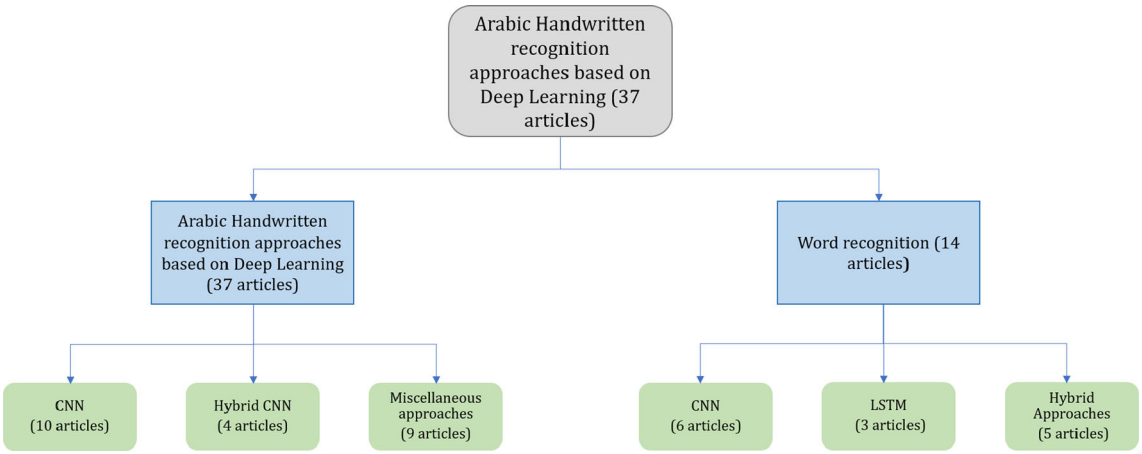


Fig. 5 Categories of Arabic handwritten recognition in the state-of-the-art

5 State-of-the-art in AHR Based on DL Approaches

As illustrated in Figure 5, the state-of-the-art in Arabic Handwritten Recognition can be broadly categorized into two categories based on the type of handwritten dataset:

- The first category comprises the approaches that used isolated handwritten characters datasets.
- The second category includes works that used the datasets of Arabic handwritten words and sub-words.

Different deep learning techniques have been used in developing recognition systems for isolated Arabic Handwritten characters. They are mainly categorized into three categories: CNN-based approaches, CNN-based hybrid approaches, and miscellaneous techniques. They can be categorized as CNN approaches, LSTM approaches, or hybrid approaches to recognize handwritten words.

5.1 Handwritten Arabic Character Recognition.

5.1.1 CNN Based Approaches for AHCR

This section reviews the works reporting on typical CNN architectures that involve alternating layers of convolutional, activation, and pooling layers for feature extraction, followed by SoftMax FCL for classification.

El-Sawy et al. [19] claimed that CNN approaches outperform other methods in feature extraction and classification tasks, and they are suitable for use in Arabic character recognition models. However, they require a large dataset to achieve good results, while the available Arabic Handwritten Character datasets have a limited number of images. Therefore, they developed the Arabic Handwritten Characters Dataset (AHCD) and designed a CNN model. The CNN approach provides an accuracy of 94.9%. Similar to the previous study, Altwaijry et al. in [2] introduced the Hijja dataset, a new Arabic handwritten character dataset, with characters written by children aged from 7 to 12 years old. They also proposed a CNN-based model for recognizing handwritten Arabic characters.

To evaluate their model, they compared its performance with the one introduced in El-Sawy's paper [19] on both AHCD and Hijja datasets. From the empirical results, Altwaijry's model outperformed the other model with an accuracy of 88% and 97% on the Hijja and AHCD datasets, respectively, while the other model achieved an accuracy of 80% and 93.84% on the Hijja and AHCD datasets, respectively. Balaha et al. in [38] introduced an Arabic handwritten character dataset (HMDB) and two CNN-based architectures, HMB1 and HMB2. The former has a more significant number of hidden layers and trainable parameters to investigate the impact of architecture complexity in recognizing Arabic handwritten script. The two proposed architectures were trained and tested on three datasets: HMBD, CMATER, and AIA9k. Through 16 experiments, the effect of data augmentation, regularization, weight initializers, and optimizers on the models' performance was studied. The best results were 90.7 %, 97.3%, and 98.4% for HMBD, CMATER, and AIA9k, respectively. They compared their architecture with the one proposed in [19] by training and testing it on HMBD, but the accuracy of the model did not exceed 47.7%.

With the aim of enhancing the results achieved on AHCD, both Younis [20] and Najadat et al. [29] designed CNN models that were trained and evaluated on AHCD. The model of [20] was composed of three convolutional layers and a fully connected layer. Also, two regularization techniques, Dropout and Batch Normalization, with different data augmentation methods were utilized to improve the model's performance. The model was trained and evaluated using AIA9k and the AHCD datasets and the achieved accuracy

was 94.8% and 97.6%, respectively. At the same time, the CNN architecture proposed in [29] consists of four convolutional layers and two max-pooling layers, followed by three fully connected layers. They tested different epochs and batch values and found that the best results were achieved when 40 epochs trained the model with a mini-batch equal to 16. From the empirical results, the model accuracy reached 97.2%.

VGG16 [57] is a very deep CNN architecture built to investigate how the network depth affects its accuracy in a large-scale image recognition setting. It consists of 16 layers and is used in different handwriting recognition applications. De Sousa [6] and Mudhsh [21] designed models for Arabic handwritten character recognition inspired by VGG-16. De Sousa in [6] proposed two deep CNN-based models, VGG-12 and REGU, for recognizing Arabic handwritten characters and digits. VGG-16 inspired the VGG-12 model by removing the fifth convolutional block and adding a dropout layer before the SoftMax FCL that is used as a classifier. In contrast, the REGU model was designed from scratch by adding different dropout and batch normalization layers at both the CNN and the fully connected layers.

The two models were trained once without data augmentation and once with data augmentation. Then, an ensemble model of the four models was created by averaging the predictions of each of the four models. The best Test accuracy of the ensemble model was 99.47%. Mudhsh et al. in [21] proposed a model for Arabic handwritten digits and character recognition. The model consists of thirteen convolutional layers, two max-pooling layers, followed by three fully connected layers. To reduce model complexity and training time, only one-eighth of the filters in each layer of the original VGG-16 were used in the proposed model. Two different datasets were used to train and evaluate the model: ADBase was used for the digit recognition task, while HACDB was used for the character recognition task. To avoid the overfitting problem, they used dropout and data augmentation. The achieved accuracy of the model was 99.66% using the ADBase database and 97.32% for the HACDB database.

Boufenar et al. in [5] used AlexNet, a popular CNN architecture that consists of 5 convolutional layers and three max-pooling layers followed by three fully connected layers. They studied the effect of preprocessing in improving the model results. The AlexNet model was trained and evaluated in two handwritten character datasets, AHCD and OIHACDB-40, with three learning strategies: training the CNN model from scratch, using a transfer learning strategy, and fine-tuning the CNN. The experimental results show that the first strategy achieved higher results, with 100% and 99.98% accuracies for OIHACDB-40 and AHCD, respectively.

Alyahya et al. in [37] investigated the performance of ResNet-18 architecture in recognizing handwritten Arabic characters when FCL and Dropout were added to the original architecture. They designed four deep models: two models that used a Fully Connected Layer with/without a dropout layer after all convolutional layers and two models that used two Fully Connected Layers with/without a dropout layer. They used the AHCD dataset to train and evaluate the CNN-based ResNet-18 model, and the best test result was 98.30%, achieved by the original ResNet-18.

Almansari et al. [30] compared the performance of CNN with multilayer perceptron (MLP) in recognizing Arabic characters of the AHCD dataset. To reduce the overfitting of the models, they tested different dropout values, batch size, number of neurons in the MLP model, and filter size in the CNN model. Per the experimental results, 95.3% and 72.08% were the best test accuracies achieved by the CNN model and the MLP model, respectively, indicating that CNN models are more suitable for Arabic handwritten character recognition.

Musa [74] presented a paper that described the efforts of the SUST-ALT (Sudan University of Science and Technology- Arabic Language Technology) research group towards building datasets for research in recognition of Arabic handwritten. The datasets contain numerals datasets, isolated Arabic letters datasets, Arabic names datasets. Most of these datasets are offline. The paper also describes some published results as well as future work

Noubigh et al. [75] investigated the problem of handwritten Arabic recognition. They propose a new architecture combining CNN and BLSTM based on a character model approach with a CTC decoder. The handwriting Arabic database KHATT is used for experiments. The results demonstrate a net advantage of performance for the CNN-BLSTM combining approach compared to the approaches used in the literature.

From the papers above, it can be seen that models designed from scratch yielded better results since pre-trained models are trained on a general dataset and might not necessarily be suited for Arabic handwriting classification. Pre-trained models are machine learning and deep learning models designed and trained on a wide variety of data. For example, the ImageNet dataset contains thousands of images and is used to train models. However, the models developed here used the image dataset to start the training of the models from the very beginning making them a lot more suitable for the classification task since they thoroughly understand the dataset. Then the deep learning models were altered by adjusting the hidden layers making them more suited for the classification task.

Table 11 summarizes the details of CNN-based approaches for recognizing isolated Arabic handwritten char-

Table 11 Previous works done on Arabic handwritten recognition showing the dataset used, AI methods used, and how the models were optimized (controlling the parameters of the models to achieve the best accuracies)

Researcher(s)	Year	Dataset	Method	Regularization method (to prevent overfitting)	Results (in Accuracy)
El-sawy et al. [19]	2017	AHCD [19]	CNN	Mini-batch	94.9%
Altwaijry et al. [2]	2020	AHCD [19]	CNN	Dropout Weight Decay	97%
		Hijja [2]			88%
Younis [20]	2017	AHCD [19]	CNN	Dropout Batch Normalization Varying Learning Rate Data Augmentation: translation, rotation, adding Gaussian noise	97.6%
		AIA9k [48]			94.8%
Najadat et al. [29]	2019	AHCD [19]	CNN	Dropout Batch normalization	97.2%
De Sousa [6]	2018	AHCD [19]	CNN (based on VGG)	Dropout Batch Normalization Data Augmentation: zoom, translation ReduceonLRPPlateau: function for swapping between SGD and Adam optimizer	98.42%
Mudhsh et al. [21]	2017	HACDB	CNN (based on VGG)	Dropout Data Augmentation	97.32%
Boufenar et al. [5]	2017	AHCD [19]	CNN (Alexnet)	Dropout Mini-batch	100%
		OIHACDB-40			99.98%
Alyahya et al. [37]	2020	AHCD [19]	CNN		98.30%
Almansari et al. [30]	2019	AHCD [19]	CNN and MPL	Dropout	95.27% (CNN) 72.08%. (MLP)
Balaha et al. [38]	2020	HMBD [38]	CNN	Dropout	90.7%
		AHCD [19]			97.3
		AIA9k			98.4%
Musa [74]	2011	SUST-ALT dataset	CNN pattern recognition		Very high accuracy
Noubigh et al. [75]	2021	KHATT	CNN and BLSTM	CTC decoder	Very high accuracy

acters showing the methods used for the classification by various researchers and the optimization techniques they used to get the best and most effective model. The best recognition accuracy reported so far for isolated handwritten character recognition is 100% [5]. This recognition accuracy was obtained by training AlexNet from scratch with the AHCD dataset.

5.1.2 Hybrid CNN Approaches

This section reviews works reporting on Hybrid CNN architectures that involve alternating layers of convolutional, activation, and max-pooling layers for feature extraction. The resulting feature maps are used as input to a conventional ML algorithm, e.g., SVM or K-mean clustering, to carry out the classification task.

Elleuch et al. [12] claimed that CNN achieves excellent results in terms of feature extraction; however, they used fully connected layers (FCLs) as the last layer to perform the classification task. Elleuch et al. in this paper, claimed no theoretical relationship between FCL that uses MLP and classification task based on different studies. On the other hand, SVM achieved excellent results in improving the performance of classification tasks. Therefore, they designed a model that used CNN for feature extraction and SVM as a classifier. They used the HACDB dataset to train the model. The model achieved a 6.59% Error Classification Rate (ECR), whereas the standard CNN model achieved an ECR of 14.71%. Also, Elleuch et al. in [13] improved the previous model by adding the dropout technique, and the ECR was reduced to 5.83%.

Elleuch et al. [31] designed a hybrid model for recognizing Arabic words or characters consisting of a Convolutional

Deep Belief Network (CDBN) for feature extraction and an SVM for classification. To reduce overfitting, the dropout technique was added to the model. They used HACDB, a dataset of isolated letters, to train a model to recognize isolated characters, and IFN/ENIT, a dataset of words, was used to train the model on whole words. For evaluating the first model, the character error rate (CER) was used as the performance metric, while the word error rate (WER) was used to evaluate the second model. Before dropout, the results for HACDB were 1.82% CER, and after dropout, the CER was reduced to 1.14%. The results for IFN/ENIT were 8.45% WER before dropout and 7.1% after applying dropout. Table 11 shows a comparison between the hybrid models of Arabic handwritten character recognition discussed above.

Shams et al. [39] proposed a deep learning model composed of CNN with FCL; then, the SVM was used to boost the results of the FCL. They focused on solving the problem of multi-stroke Arabic character recognition, i.e., recognition of letters that have the same stroke but differ in the number and position of the dot(s), by using an unsupervised K-mean clustering algorithm to group the characters into 13 clusters. They used AHCD [19] to training and evaluate the model. The model achieved 95.07% in terms of accuracy, outperforming the Al-sawy model, which achieved 94.90%.

Theoretically, hybrid models should have the best accuracy since they combine traditional machine learning and deep learning. This is due to the fact that hybrid models combine the best of both worlds; deep learning models are great for feature extraction, while machine learning models are great for classification. All the experiments performed above aimed to use deep neural networks to extract the right features that will be used to distinctly classify the categories and pass on these features to the machine learning models which use these features for classification. However, in the papers reported above, the accuracy is lower than those of standalone models, resulting from not understanding how to fine-tune the hyperparameters of both models and/or the deep neural network not extracting the right features passed on to the machine learning models. In addition, with more and more experiments being conducted, researchers hope to design hybrid models that will surpass all standalone models correctly.

The details of hybrid approaches for recognizing isolated Arabic handwritten characters are summarized in Table 12.

5.1.3 Miscellaneous Techniques

This section reviews works on different approaches that can be considered deep learning architectures but do not use CNN for feature extraction.

- Fusion of Classifiers and Handcrafted Feature Extraction Methods

Ali et al. in [32] assumed that applying fusion strategy at both feature extraction and classification levels can improve the performance of Arabic handwritten recognition systems. Therefore, they designed a model that consisted of four feature extraction methods, namely moment invariants (MI), run-length matrix (RLM), statistical properties of intensity histogram (SFIH) and wavelet decomposition (WD); and three classifiers, modified quadratic discriminate functions (MQDF), support vector machine (SVM) and random forest (RF). Both AHCD and IFN/ENIT datasets were used to test the model. From the experiment results, these fusion strategies scored better results than traditional methods.

- SVM + ANN

Al-Jubouri, in his master thesis [22], designed a new hybrid model for recognizing isolated Arabic characters. SVM is used in the first stage to divide the characters into two groups: characters without dot(s) and characters with the dot(s). Then a Neural Network (NN) is used to complete the classification process. The IFN/ENIT dataset was used to train the model. It consists of Arabic words and isolated characters, but the model was trained only on the isolated characters. The Error rate was used as the evaluation metric. The SVM+NN achieved a 7.8% error rate, while the error rate of the NN model was 28.05%. The author mentioned that the model suffered from a vanishing gradient. Similar to the previous study, Al-Jourishi et al. [40] introduced a hybrid architecture that consisted of SVM and NN. They used the AlexU Isolated Alphabet (AIA9K) dataset to train the model. Their

Table 12 A table showing the hybrid approaches in Arabic handwritten character recognition with year, the dataset used, the methodology applied, the evaluation metric (how the effectiveness of the models was tested), and the results obtained

Paper	Year	Dataset	Hybrid model architecture	Evaluation metric	Results
Elluech et al. [12]	2016	HACDB	CNN + SVM	ECR	6.59%
Elluech et al. [13]	2016	HACDB	CNN + SVM	ECR	5.83%
Elluech et al. [31]	2019	HACDB	CDBN + SVM	CER	1.14%
		IFN/ENIT			7.1%
Shams et al. [39]	2020	AHCD [19]	CNN + FCL + SVM	(ACC)	95.07%
				(ECR)	4.93%

approach achieved an accuracy of 99.14% in the first stage and 91.84% in the second stage.

- Deep SVM

Elleuch et al. investigated the use of SVM in improving Arabic handwriting recognition in different papers. In [14, 15] they designed a deep model called Deep SVM (DSVM) that consisted of a stack of SVM layers and compared it with models that used handcrafted feature extractors, i.e., SIFT descriptor [14] and Gabor filter [15]. Both models were trained on HACDB, which consists of 6600 images of different Arabic character shapes. To improve the DSVM model, they perform a grid search for parameter tuning and added dropout to reduce overfitting. They used the error classification rate as the evaluation metric. The DSVM outperformed models that used handcrafted feature extraction, with a 5.68% error classification rate.

- Deep Belief Network

Among the earlier works, Elleuch et al. investigated the effectiveness of using Deep Belief Neural Networks (DBNN) in recognizing the Arabic handwritten script. In [10], they evaluated DBNN in recognizing handwritten characters and words. The DBN architecture consisted of an input layer, 2 Restricted Boltzmann Machine (RBM) layers, and an output layer. The performance of DBNN was evaluated on two datasets: HACDB and ADAB, which consist of words and are used for online handwritten recognition. The model achieved a classification error rate of 2% and 41% for the HACDB dataset and ADAB dataset, respectively. They concluded that DBNN is useful for character-level recognition but not efficient in online handwriting recognition when the size of images is dynamic. In [11], they compared CNN and DBNN in recognizing 24 and 66 shape characters of HACDB. They used the same DBN architecture proposed in [10]. The CNN architecture consisted of an input layer, two Convolutional layers, and two pooling layers, followed by a fully connected layer and an output layer. The DBN achieved a 1.67% and 3.64% error classification rate (ECR) for 24 classes and 66 classes of HACDB, respectively, outperforming the CNN, which reported 5% and 14.71% ECR for 24 classes and 66 classes of HACDB, respectively.

Dropout and DropConnect are two famous regularization techniques used to improve deep learning models by reducing overfitting. In [23], Elleuch et al. studied their effect by applying them in the DBN architecture proposed in [11]. The system achieved an error rate of 2.73% and 2.27% after applying Dropout and DropConnect, respectively, reducing the error rate of the original DBN architecture by 1.37%. The previous work was extended in [26] by applying Dropout and DropConnect in both DBN and CNN architectures presented in [11]. They applied the Dropout and DropConnect techniques during the training on each architecture. The best result reported by DBN was a 2.27 % error rate when DropConnect was applied, while the best result reported by CNN was 3.79 % when Dropout and DropConnect were jointly used.

Theoretically, hybrid models should have the best accuracy due to the fact that they combine traditional machine learning and deep learning. This is due to the fact that hybrid models combine the best of both worlds; deep learning models are great for feature extraction, while machine learning models are great for classification. All the experiments performed above aimed to use deep neural networks to extract the right features that will be used to classify the categories distinctly and on these features to the machine learning models which use these features for classification. Nevertheless, in the papers reported above, the accuracy is lower than those of standalone models, resulting from not understanding how to fine-tune the hyperparameters of both models and/or the deep neural network not extracting the right features to be passed on to the machine learning models. The models trained here are experimental and still not well understood, which is why they have relatively lower accuracies as compared to the standalone models. More experiments are needed with a variation of hyperparameters so as to understand the processes better.

The details of miscellaneous approaches for recognizing isolated Arabic handwritten characters are summarized in Table 13.

5.2 Arabic Handwritten Words Recognition

5.2.1 CNN Architectures

Ghanim et al. in [41] proposed a new approach that aims to reduce the Arabic script recognition complexity while improving the recognition accuracy by mining the huge datasets before classification. The approach consists of three stages: matching, ranking, classifying. The first and second stages reduced the dataset size to be only 11% of the dataset. At the classification stage, they compared the performance of SVM and different well-known Deep CNN (DCNN) architectures, e.g., AlexNet VGG-16 and GoogleNet. They showed that the performance of the AlexNet is improved after applying the matching and ranking stages, there are 946 classes in IFN/ENIT before matching and ranking, and that number is reduced to 100 classes. Also, the performance is improved from 76% in terms of accuracy and 72.1 minutes for training to 95.6% and 18.5 minutes for training. The best result of SVM was 14.6% in WER. They compared between the most popular CNN architectures: Residual Networks (ResNets) [58], AlexNet architecture [59], VGG-16 architecture [60], GoogleNet architecture [61], ResNeXT [62],

Table 13 Miscellaneous techniques in Arabic handwritten character recognition with the years, the dataset used, the architecture or type of model used, the evaluation metric, and the performance result

Paper	Year	Dataset	Architecture	Evaluation metric	Performance
Ali et al. [32]	2019	AHCD + IFN/ENIT	Fusion of classifiers	–	–
Al-Jubouri [22]	2017	Characters in IFN/ENIT	SVM + NN	Error rate	7.8% (the first stage) 28.05% (the second stage)
Al-Jourishi et al. [40]	2020	Dataset	Architecture	Accuracy	99.14% (the first stage) 91.84% (the second stage)
Elluch et al. [14]	2016	AIA9K	SVM + NN	ECR	5.68%
Elluch et al. [15]	2016	HACDB	Deep SVM	ECR	5.68%
Elluch et al. [10]	2015	HACDB	Deep SVM	–	2.1% error classification rate
		HACDB (66 characters shapes)	Deep Belief Neural Network (DBNN)	–	40% error rate
Elluch et al. [11]	2015	ADAB (online words)	HACDB (24 characters shapes)	ECR	1.67%
		HACDB (24 characters shapes)	DBN		3.64%
		HACDB (66 characters shapes)	DBN		5%
		HACDB (24 characters shapes)	CNN		14.71%
Elluch et al. [23]	2017	HACDB (66 characters shapes)	CNN DBN with Dropout	Error rate	2.73%
		HACDB			2.27%
Elluch et al. [26]	2018	DBN with Dropout	DBN with Dropconnect	Error rate	2.73%
		HACDB			2.27%
		DBN with Both Dropout and Dropconnect			2.65%
		CNN with Dropout			4.62%
		CNN with Dropconnect			4.32%
		CNN with Both Dropout and Dropconnect			3.79%

and DenseNet [63]. The AlexNet achieved the best result with an accuracy of 95.6%.

Poznanski and Wolf in [16] designed a novel framework for handwritten word recognition adaptable for different languages. In their framework, instead of using the holistic approach to recognize the word as one unit, the system uses attributes-based encoding, in which the image is described as having or lacking a set of binary attributes. An example of binary attributes is "does the first half of the word has the letter A." A CNN network that VGG inspires was used, and rather than using one fully connected layer, they used multiple separate and parallel fully connected layers. Each layer leads to a separate group of attributes predictions. Dropout, weight decay, and data augmentation, namely rotation and shear, were used to reduce overfitting. They trained and tested the system on three handwriting recognition benchmarks, namely the IFN/ENIT, IAM, and RIMES for Arabic, English, and French, respectively. The system achieved the state-of-the-art results with 97.07% on IFN/ENIT dataset.

Almodfer et al. [24] used the Multi-Column Deep Neural Network (MCDNN) [64] to recognize Arabic handwritten words. MCDNN architecture is represented as columns, and each column is a CNN. The CNNs have the same architecture and are trained on the same data. However, the data are preprocessed differently in each of the CNNs the outputs of the networks averaged. They built an MCDNN with three columns (CNN) and trained and tested on IFN/ENIT and the best result was an 8.5% error character rate.

El-Melegy et al. in [33] investigated the performance of CNN in recognizing Arabic handwritten literal amounts that are used in bank cheques. Also, they studied the effect of data augmentation in improving the model performance. Therefore, a CNNs architecture of 17 layers is designed, trained and evaluated on AHDB [53]. They used 80% of the dataset for training and 20% for testing. The model achieved 96.58% and after applying data augmentation, the model achieved 97.8%.

Mustafa and Elbashir in [42] proposed a CNN architecture to recognize the male names in the dataset [54]. They used batch normalization and dropout to improve the model performance. The model achieved an accuracy of 99.14%. The authors illustrated that the dataset has only 20 classes, and the accuracy may reduce if the number of classes is increased.

Almodfer et al. in [25] claimed that the main cause of overfitting in deep CNN models in Arabic handwritten recognition is the small size of the datasets. Therefore, they investigated how the dropout regularization technique can alleviate the impact of overfitting. They designed four models based on AlexNet [59] by adding dropout [65] at different positions and values. They also studied which activation function is more appropriate for the FCLs: ReLU or Tanh. The imperial results showed that applying incremental dropout values improves the model performance. Also, the activation functions achieved comparable results with accuracies of 92.13% and 92.55% for ReLU and TanH, respectively.

Alsaffar et al. [73] presented a paper highlighting the preprocessing and binarization methods used in the literature and proposed numerous directions for developing. They reviewed the various current deep learning approaches and tools used for Arabic handwritten recognition (AHWR), identified challenges along this line of this research, and gave several recommendations, including a framework based (DL) that is particularly applicable for dealing with cursive nature languages

From the papers above, it can be seen that CNN networks designed were properly fine-tuned and variation of hidden layers showed which gave the best results. The activation functions were also varied with the aim of finding the best for the job. The model's performance was satisfactory. The details of CNN approaches for recognizing Arabic handwritten words are summarized in Table 14.

5.2.2 LSTM Architectures

Maalej and Kherallah in [17] claimed that RNN and its variations, e.g., LSTM and MDLSTM are the most successful approaches to be applied in word recognition when using a holistic approach. They introduced a new Arabic handwriting word recognition architecture based on Multi-Dimensional LSTM (MDLSTM). However, MDLSTM tends to suffer from overfitting that results from the huge number of the parameters. Therefore, they studied how the dropout regularization technique reduces the model overfitting and improves its performance. The performance of the basic MDLSTM was 16.97% in label error rate and it was improved to be 12.09 % after applying dropout. Since the dropout significantly improves the performance of the model, in [18], they investigated the performance of the same MDLSTM when dropout is applied at different positions, e.g., before, after or inside the MDLSTM layers. The empirical results proved that whenever dropout is applied, it reduced the error rate and the best result was 11.62 % in label error rate achieved when dropout is applied before the MDLSTM layer.

Maxout is another regularization technique that was investigated by Maalej and Kherallah in [34]. They used the model proposed in [17] and implemented the Maxout at different positions: Inside LSTM units as activation function and at the feedforward layers instead of the sigmoid functions. For each architecture, a different value of Maxout group size was tested. The empirical results proved that the Maxout significantly reduce the error rate compared to the original architecture. The best result was 10.11% in label error rate that achieved when Maxout units were added to feedforward layers with the group size equal to 4.

Alkhawaldeh [76] proposed an adapted deep hybrid transfer model developed using two well-known pre-trained convolutional neural networks (CNN) models. These are further adapted by adding recurrent neural networks, especially long short-term memory (LSTM) architectures to detect Arabic (Indian) Handwritten Digits (AHD). The CNN model learns the relevant features of Arabic (Indian) digits, while the sequence learning process in the LSTM layers extracts long-term dependence features. The experimental results, using popular datasets, show significant performance obtained by the adapted transfer models with accuracy reached up to 98.92% and precision and recall values at most cases reached 100% with statistical tests using p-value ($p \leq 0.05$) compared to baseline methods.

From the papers reported, it is seen that RNN models generally performed poorly compared to CNN models. This is due to the fact that CNN performs a better job when it comes to learning to recognize patterns across space, and it is most suitable for spatial data (limited data). The RNN models will work better in the future when more data has been gathered for Arabic handwritten words. The details of LSTM approaches for recognizing Arabic handwritten words are summarized in Table 15.

5.2.3 Hybrid Architectures

Ahmad et al. in [43] claimed that the performance of Arabic recognition models is affected by the limited size of the Arabic dataset. Therefore, they investigated the capability of data augmentation methods to improve the performance of deep learning models on the KHATT dataset. They implemented a model using (MDLSTM) and used Connectionist Temporal Classification (CTC) for classification. Four different data augmentation methods were applied to enrich the dataset: blurring, text-line contouring, edge enhancing, and the bleed-through effect. The model achieved 19.98% in terms of label error rate and 4.22% as character error rate.

Table 14 CNN approaches in Arabic word character recognition

Paper	Year	Dataset	Method	Data enhancement	Results
Ghanim et al. [41]	2020 IEEE	IFN/ENIT Arabic database (words)	CNN (AlexNet)	Matching, ranking,	95.6% (Accuracy)
Poznanski and Wolf [16]	2016	IFN/ENIT	CNN—multiple separate and parallel fully connected layers	Dropout applied at fully connected layers and Weight decay data augmentation [rotation, shear,	97.07% (Accuracy)
Almodfer et al. [24]	2017	IFN/ENIT	Multi-Column Deep Neural Networks (MCDNN) *they refer to CNN as DNN	–	8.5% error rate
El-Melegy et al. [33]	2019	AHDB	CNN	Data augmentation	97.8% (Accuracy)
Mustafa and Elbashir [42]	2020	SUST-ARG names [54]	CNN	Batch normalization and dropout	99.14% (Accuracy)
Almodfer, Rolla, et al. [25]	2017	IFN/ENIT	CNN (AlexNet + ReLU)	dropout	92.13% (Accuracy)
			CNN (AlexNet + Tanh)		92.55% (Accuracy)

Table 15 LSTM approaches in Arabic word character recognition

Paper	Year	Dataset	Method	Data enhancement and optimization	Results
Maalej and Kherallah [17]	2016	IFN/ENIT	MDLSTM	Dropout Early-stopping	12,09% (label error rate)
Maalej and Kherallah [18]	2016	IFN/ENIT	MDLSTM	Dropout	11.88% (Label error rate)
Maalej and Kherallah [34]	2019	IFN/ENIT	MDLSTM	Maxout	10.11% (error rate)
Alkhawaldeh [76]	2021	IFN/ENIT	MDLSTM		98.92% accuracy

Maalej and Kherallah [27] evaluated different hybrid models proposed in Arabic Handwritten word recognition. It is argued that CNNs are designed to learn local higher-level features from spatial data; however, they cannot detect sequential correlations. On the other hand, BLSTM are designed and used for sequential modeling; however, they require high runtime and memory for training and the location activations during back-propagation. To alleviate these drawbacks, they proposed a new model in [66] that uses CNN for automatic feature extraction and BLSTM with CTC for sequence labeling. Dropout and data augmentation, namely tilts, skew, shears, horizontal and vertical elongation and shortening, were used to reduce network overfitting. The model was trained and tested on IFN/ENIT and achieved an accuracy of 92.21%.

Khemiri et al. in [35] discussed that Bayesian Network (BN) architectures and Convolutional Neural Network (CNN) are different types of ML in the process of defining the distinctive features of the dataset. BN needs the domain experts to manually define the features to be used as input to the NB algorithm. While CNNs learn the feature automatically by training the model on the training data. Therefore, they investigated the performance of these two architectures and combinations of the architectures in recognizing the images of Arabic words in IFN/ENIT. They train and evaluate five BN architectures e.g., Naïve Bayes (NB) and Dynamic and Naïve Bayes (DNB) and 6 CNN architectures. The DNB model achieved the best performance in the BN category with accuracy of 88.27%. The best result achieved by the CNN model was 92.7%. The combination of the previous best models achieved an accuracy of 95.20%, improving the performance of the original two models

Amrouch et al. in [28] investigated the performance of HMM when the hand-coded features are replaced with the features that are automatically extracted by CNN in the field of Arabic Handwritten recognition. They study the effect of the dataset size in the performance of the hybrid CNN-HMM models. They used the IFN/ENIT dataset which consists of 5 subsets: a, b, c, d, and e. In the experiment, they used two train and test scenarios: (abc-d) scenario: where the models are trained with a, b, c subsets and evaluated in e subset, and (abcd-e) scenario where the model trained on the first four subsets and evaluated one subset. The HMM model achieved 87.93 % in terms of accuracy. While the CNN-

Table 16 Hybrid Approaches in Arabic handwritten word recognition

Paper	Year	Dataset	Method	Data enhancement and optimization	Results
Ahmad et al. [43]	2020	KHATT	MDLSTM + CTC	Data augmentation	19.98% (label error rate) 4.22% (character error rate)
Maalej and Kherallah [27]	2018	IFN/ENIT	(hybrid) CNN-LSTM—CTC	Dropout Data augmentation (tilt, skew, shears, h and v elongation and shortening)	92.21% (Accuracy)
Khémiri et al. [35]	2019	IFN/ENIT	DNB		88.27 (Accuracy)
			CNN		92.7 (Accuracy)
			CNN + DBN		95.20 (Accuracy)
Amrouch et al. [28]	2018	IFN/ENIT (abc-d)	HMM		87.93 (Accuracy)
			CNN—HMM		88.95 (Accuracy)
		IFN/ENIT (abcd-e)	CNN—HMM		89.23 (Accuracy)
Awni et al. [36]	2019	IFN/ENIT	ResNet18 with Adam		7.21% (error rate percentage)
			ResNet18 with SGD		9.41% (error rate percentage)
			ResNet18 with RMSProp		11.57% (error rate percentage)
			Ensemble of three ResNet18 models		6.63% (error rate percentage)

HMM model achieved accuracies of 88.95% and 89.23% in the first and second scenarios. This data shows that the performance of deep models is improved by increasing the dataset size.

Awni et al. in [36] investigated the applicability of ensemble models in recognizing handwritten Arabic words. The ensemble model is the optimal model obtained by integrating different models together. They used three different optimization techniques: RMSprop, SGD and Adam, to train three ResNet18 models. They compared the results of the individual models with the results obtained by the ensemble of these models. The obtained results in terms of error rate were 11.57%, 9.41%, and 7.21 %, for RMSprop, SGD, and Adam, while the ensemble model improved the base model results achieving 6.63% in terms of error rate.

As mentioned above, the hybrid models trained are in their experimental stages and hence produce a lot of irregularities and not so effective result. Nevertheless, as more and more experiments are conducted, there is hope for a massive improvement in the efficiency of the models. The details of hybrid approaches for recognizing Arabic handwritten words are summarized in Table 16.

6 Analysis of the State-Of-The-Art

The analysis of the State-Of-The-Art approaches cited in this survey are discussed. This includes a detailed investigation of the pros and cons of the approaches, their strengths and weaknesses, which approaches were successful and how they were evaluated.

A very crucial part of deep learning that could affect all the State-Of-The-Art is the quality and size of the dataset used as this is the benchmark that guarantees effective results. Insufficient and low-quality data will lead to ineffective and inconsistent results in all the State-Of-The-Art techniques employed. Also, Arabic has a few characteristics which make it difficult to train machine learning models. Such characteristics include: the cursive nature of Arabic (cursive is a style of penmanship in which the symbols of the language are written in a conjoined and/or flowing manner), its grammatical complexity, the presence of dots and dashes, among others, makes Arabic handwritten recognition a very complicated and challenging problem.

There are two main types of metrics used for machine learning classification problem, runtime metrics which performs an assessment on the classification models. At the same time, they are being trained and used, and evaluation metrics which performs assessments on the classification models after the training is done through testing, for example, getting

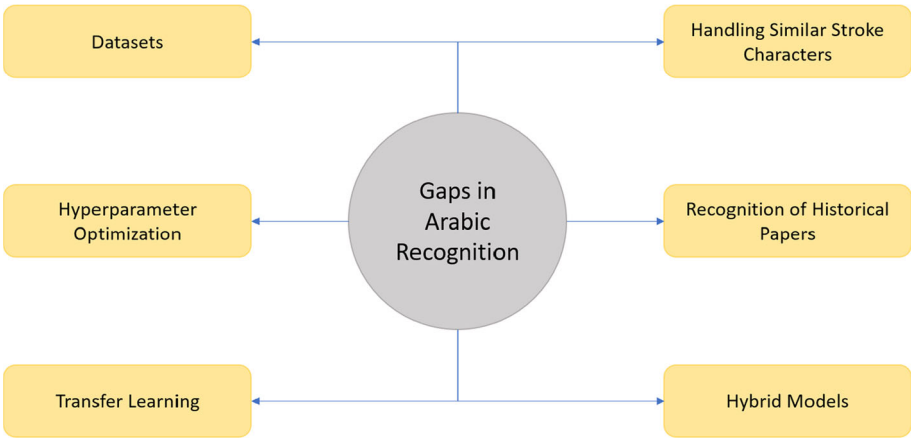


Fig. 6 Challenges and Open Problems encountered in AHR generally

the accuracy and error rate. However, the State-Of-The-Art approaches discussed in this survey mainly report evaluation metrics, so it is our main focus.

Due to lack of sufficient data of Arabic handwritten images, the models to be trained by all the State-Of-The-Art were prone to overfitting, which happens when a model learns the details and noise of training data to the extent that it negatively impacts the performance of the model on new data. For the State-Of-The-Art techniques, a few methods were used in the studies discussed to mitigate the impact of overfitting. These methods include: dropout [67], batch normalization [68], Maxout [69], weight decay [70] and Dropconnect [71] and data augmentation. The experiments analyzed show that using more datasets for training yielded better results. For example, when the two results gotten in [28] were compared, it was seen that the experiment that was conducted on the abcd subsets of the IFN/ENIT dataset yielded better results as compared to the experiment conducted on the abc subset of the dataset, even though the same model was used. This goes to show us that in machine learning, the more the dataset available for training, the better and more effective the results obtained will be, and it also reduces the chances of the classification models to overfit.

The high recognition rates in AHCR are achieved using very deep CNN models that apply regularization techniques: dropout, batch normalization, and data augmentation. These deep CNN models yield better results because deep networks have more explaining power as compared to shallow networks. After all, it has more hidden layers. The downside to building a deep CNN is that if there is not enough dataset for training, the desired output will not be begotten, and the network will be ineffective. In the hybrid models, however, SVM was used as a classifier in different AHCR models, while CTC, DNB and HMM were used as classifiers in AHWR. Several studies have been done in the field of Handwritten Arabic scripts, and most of them reported satisfying results in recognizing Arabic characters. Even though the dataset available for the experiments is relatively low, techniques such as data augmentation are employed to facilitate the handwritten scripts experiments.

Normally, RNN approaches such as LSTM are used for word recognition because they use sequential data and perform classification based on the segmentation of the words. However, the state-of-the-art techniques discussed in AHWR used holistic approaches. From the findings, it was observed that the CNN-based approaches yielded better results, and the models designed had better efficiencies when compared to the RNN based approaches. It is also seen from the summarization tables that AHCD and IFN/ENIT are the most commonly used datasets in AHCR and AHWR, respectively, and the state-of-the-art models designed mainly used CNN approaches, thereby yielding higher and more effective results.

In conclusion, the studies above show that the effectiveness of a machine learning model for classification is highly dependent on the dataset used and how the dataset is prepared before experiments start. It is also observed that deep learning models that were created from scratch took longer time and more computational power to train but yielded far more accurate and effective results. In contrast, models that used pre-trained models have had an average result due to the fact that the pre-trained models were trained on a general large scale image dataset such as ImageNet. The characteristics of the latter are totally different and not necessarily adaptive to AHCR and AHWR. Also, optimization of the hyperparameters played a major role because the models have to be fine-tuned and arranged in such a way that it yielded satisfying results.

7 Challenges and Open Problems

There are many challenges for researchers to work on in this field. Figure 6 shows gaps and challenges that should be tackled by future researches so as to have more effective models. There is a demand for new methods to emerge as

computational technology is increasing and resource limitations decrease. These challenges constitute the reasons why AHCR and AHWR experiments yield not so effective results. They are discussed in detail below.

The main issue faced with the majority of research in the fields of AHCR and AHWR is the lack of specific application dataset. As mentioned earlier, the size and quality of the dataset used for an experiment greatly enhance the results that will be obtained, but the available databases and repositories are limited, and, in some cases, the available ones are not of good quality.

Another problem researchers face is the inconsistency of the Arabic language with characteristics such as similar stroke characters. Dots also play an important role in differentiating between the Arabic characters that have the same stroke. Indeed, The Arabic handwritten character datasets use an image size of 32*32 pixels or 28*28 pixels, and the dots may be lost or deleted due to low resolution or during the preprocessing. Also, dots may be too small or misplaced due to the variability of handwriting. Characters with dots need more attention. Two-stage recognition frameworks need more investigation. These frameworks are proposed for handling the characters with dots. Where the first stage divides the characters into two groups depending on the presence or the absence of dots, resulting in a fewer number of character classes to be classified into. The second stage classifies the characters to one of the reduced numbers of classes.

Another challenge faced by researchers is the issue that arises when developing handwritten recognition for historical Arabic papers. The Arabic language has a lot of variation, especially when comparing ancient writing with modern writing. The dataset used for training is derived from modern Arabic, making all the AHWR and AHCR models virtually useless in recognizing Arabic historical papers.

After investigating the Hybrid state-of-the-art models, it is seen that Hybrid models in AHWR and AHCR significantly improve the results of base models. So, therefore, future research should focus on designing novel hybrid approaches that combine different ML methods. Also, investigating the effect of transfer learning on models designed from scratch and models that are trained on pertained models could provide a breakthrough in the field of AHWR and AHCR. Finally, the variation hyperparameters go a long way in determining how effective a classification model is. Optimization techniques should be employed to help fine-tune these models to achieve the best hyperparameters, which in turn yield highly effective results.

8 Conclusion

In this review, we have presented an exhaustive survey of the Arabic handwritten recognition approaches that used deep learning models for feature extraction. Also, we provided a detailed description of the datasets that were used to develop these models. The main objective of this paper was critically analyzing the current work in Arabic Handwritten recognition and identify the trends, problems, and challenges faced by the previous studies. Therefore, a systematic analysis has been performed with the intent to determine the efficiency of deep learning methods for feature extraction by comparing these methods based on their recognition rates. It has been shown that the CNNs approaches are the most used approach for feature extraction. The lack of a benchmark dataset was one of the most discussed issues in the literature. Based on the extensive study conducted in this article, it has been proposed that there is a need to develop hybrid feature extraction and classification approaches for achieving the best accuracy results.

Models including this one is designed to focus on either Arabic character of digits. Even the datasets available are either containing only characters or only digits. Hence, for future work, models could be designed which recognizes both characters and digits, a dataset could also be put together to contain both Arabic characters and digits thereby making the job easier for the researchers.

References

1. "Babbel Magazine." https://www.babbel.com/en/magazine/ (accessed Mar. 09, 2021).
2. N. Altwaijry and I. Al-Turaiki, "Arabic handwriting recognition system using convolutional neural network," Neural Comput. Appl., pp. 1–13, Jun. 2020, doi: https://doi.org/10.1007/s00521-020-05070-8.
3. Weldegebriel, H.T.; Liu, H.; Haq, A.U.; Bugingo, E.; Zhang, D.: A New Hybrid Convolutional Neural Network and eXtreme Gradient Boosting Classifier for Recognizing Handwritten Ethiopian Characters. IEEE Access **8**, 17804–17818 (2020). https://doi.org/10.1109/ACCESS.2019.2960161
4. Albahli, Saleh, Marriam Nawaz, Ali Javed, and Aun Irtaza. "An improved faster-RCNN model for handwritten character recognition." Arabian Journal for Science and Engineering (2021): 1–15.
5. Boufenar, C.; Kerboua, A.; Batouche, M.: Investigation on deep learning for off-line handwritten Arabic character recognition. Cogn. Syst. Res. **50**, 180–195 (Aug. 2018). https://doi.org/10.1016/j.cogsys.2017.11.002
6. de Sousa, I.P.: Convolutional ensembles for Arabic Handwritten Character and Digit Recognition. PeerJ Comput. Sci. **2018**(10), e167 (Oct. 2018). https://doi.org/10.7717/peerj-cs.167
7. R. Ahmed et al., "Offline arabic handwriting recognition using deep machine learning: A review of recent advances," in Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), Jul. 2020, vol. 11691 LNAI, pp. 457–468, doi: https://doi.org/10.1007/978-3-030-39431-8_44.
8. H. Q. Ghadhban, M. Othman, N. A. Samsudin, M. N. Bin Ismail, and M. R. Hammoodi, "Survey of Offline Arabic Handwriting Word Recognition," in Advances in Intelligent Systems and Computing, Jan. 2020, vol. 978 AISC, pp. 358–372, doi: https://doi.org/10.1007/978-3-030-36056-6_34.

9. Ali, A.A.A.; Suresha, M.; Ahmed, H.A.M.: A Survey on Arabic Handwritten Character Recognition. SN Comput. Sci. **1**(3), 1–10 (May 2020). https://doi.org/10.1007/s42979-020-00168-1
10. M. Elleuch, N. Tagougui, and M. Kherallah, "Arabic handwritten characters recognition using Deep Belief Neural Networks," Dec. 2015, doi: https://doi.org/10.1109/SSD.2015.7348121.
11. Elleuch, M.; Tagougui, N.; Kherallah, M.: Towards unsupervised learning for arabic handwritten recognition using deep architectures. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) **9489**, 363–372 (2015). https://doi.org/10.1007/978-3-319-26532-2_40
12. Kherallah, M.; Elleuch, M.; Tagougui, N.: A novel architecture of CNN based on SVM classifier for recognising Arabic handwritten script. Int. J. Intell. Syst. Technol. Appl. **15**(4), 323 (2016). https://doi.org/10.1504/ijista.2016.10000779
13. Elleuch, M.; Maalej, R.; Kherallah, M.: A New design based-SVM of the CNN classifier architecture with dropout for offline Arabic handwritten recognition. Procedia Computer Science **80**, 1712–1723 (Jan. 2016). https://doi.org/10.1016/j.procs.2016.05.512
14. M. Elleuch, R. Mokni, and M. Kherallah, "Offline Arabic Handwritten recognition system with dropout applied in Deep networks based-SVMs," in Proceedings of the International Joint Conference on Neural Networks, Oct. 2016, vol. 2016-October, pp. 3241–3248, doi: https://doi.org/10.1109/IJCNN.2016.7727613.
15. Elleuch, M.; Kherallah, M.: An Improved Arabic Handwritten Recognition System using Deep Support Vector Machines. Int. J. Multimed. Data Eng. Manag. **7**(2), 1–20 (May 2016). https://doi.org/10.4018/ijmdem.2016040101
16. A. Poznanski and L. Wolf, "CNN-N-Gram for HandwritingWord Recognition," in Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Dec. 2016, vol. 2016-December, pp. 2305–2314, doi: https://doi.org/10.1109/CVPR.2016.253.
17. Maalej, R.; Tagougui, N.; Kherallah, M.: Recognition of handwritten arabic words with dropout applied in MDLSTM. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) **9730**, 746–752 (2016). https://doi.org/10.1007/978-3-319-41501-7_83
18. R. Maalej and M. Kherallah, "Improving MDLSTM for offline Arabic handwriting recognition using dropout at different positions," in Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2016, vol. 9887 LNCS, pp. 431–438, doi: https://doi.org/10.1007/978-3-319-44781-0_51.
19. El-Sawy, A.; Loey, M.; El-Bakry, H.: Arabic Handwritten Characters Recognition using Convolutional Neural Network. WSEAS Trans. Comput. Res. **5**(1), 11–19 (2017)
20. K. Younis, "ARABIC HANDWRITTEN CHARACTER RECOGNITION BASED ON DEEP CONVOLUTIONAL NEURAL NETWORKS," 2018.
21. Mudhsh, M.A.; Almodfer, R.: Arabic Handwritten Alphanumeric Character Recognition Using Very Deep Neural Network. Information **8**(3), 105 (Aug. 2017). https://doi.org/10.3390/info8030105
22. B. Mohamed Anas Hussein Al-Jubouri, "Offline Arabic Handwritten Isolated Character Recognition System Using Support vector Machine and Neural Network إ ف لى ّة العربي الحروف المنفصلة باستخدام اليد بخط والمكتوبة الية الموجه دعم العصبية والشبكة," 2017
23. Elleuch, M.; Tagougui, N.; Kherallah, M.: Optimization of DBN using Regularization Methods Applied for Recognizing Arabic Handwritten Script. Procedia Computer Science **108**, 2292–2297 (Jan. 2017). https://doi.org/10.1016/j.procs.2017.05.070
24. R. Almodfer, S. Xiong, M. Mudhsh, and P. Duan, "Multi-column deep neural network for offline arabic handwriting recognition," in Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2017, vol. 10614 LNCS, pp. 260–267, doi: https://doi.org/10.1007/978-3-319-68612-7_30.
25. R. Almodfer, S. Xiong, M. Mudhsh, and P. Duan, "Enhancing alexnet for Arabic handwritten words recognition using incremental dropout," in Proceedings - International Conference on Tools with Artificial Intelligence, ICTAI, Jun. 2018, vol. 2017-November, pp. 663–669, doi: https://doi.org/10.1109/ICTAI.2017.00106.
26. M. Elleuch, A. M. Alimi, and M. Kherallah, "Enhancement of Deep Architecture using Dropout/ DropConnect Techniques Applied for AHR System," in Proceedings of the International Joint Conference on Neural Networks, Oct. 2018, vol. 2018-July, doi: https://doi.org/10.1109/IJCNN.2018.8489245.
27. R. Maalej and M. Kherallah, "Convolutional Neural Network and BLSTM for Offline Arabic Handwriting Recognition," Mar. 2019, doi: https://doi.org/10.1109/ACIT.2018.8672667.
28. M. Amrouch, M. Rabi, and Y. Es-Saady, "Convolutional feature learning and CNN based HMM for Arabic handwriting recognition," in Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), Jul. 2018, vol. 10884 LNCS, pp. 265–274, doi: https://doi.org/10.1007/978-3-319-94211-7_29.
29. H. M. Najadat, A. A. Alshboul, and A. F. Alabed, "Arabic Handwritten Characters Recognition using Convolutional Neural Network," in 2019 10th International Conference on Information and Communication Systems, ICICS 2019, Jun. 2019, pp. 147–151, doi: https://doi.org/10.1109/IACS.2019.8809122.
30. O. A. Almansari and N. N. W. N. Hashim, "Recognition of Isolated Handwritten Arabic Characters," Oct. 2019, doi: https://doi.org/10.1109/ICOM47790.2019.8952035.
31. Elleuch, M.; Kherallah, M.: Boosting of Deep Convolutional Architectures for Arabic Handwriting Recognition. Int. J. Multimed. Data Eng. Manag. **10**(4), 26–45 (Dec. 2019). https://doi.org/10.4018/ijmdem.2019100102
32. Ali, A.A.A.; Suresha, M.: A novel features and classifiers fusion technique for recognition of Arabic handwritten character script. SN Appl. Sci. **1**(10), 1–13 (Oct. 2019). https://doi.org/10.1007/s42452-019-1294-6
33. M. El-Melegy, A. Abdelbaset, A. Abdel-Hakim, and G. El-Sayed, "Recognition of Arabic Handwritten Literal Amounts Using Deep Convolutional Neural Networks," in Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2019, vol. 11868 LNCS, pp. 169–176, doi: https://doi.org/10.1007/978-3-030-31321-0_15.
34. R. Maalej and M. Kherallah, "Maxout into MDLSTM for offline arabic handwriting recognition," in Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), Dec. 2019, vol. 11955 LNCS, pp. 534–545, doi: https://doi.org/10.1007/978-3-030-36718-3_45.
35. Khémiri, A.; Echi, A.K.; Elloumi, M.: Bayesian Versus Convolutional Networks for Arabic Handwriting Recognition. Arab. J. Sci. Eng. **44**(11), 9301–9319 (Nov. 2019). https://doi.org/10.1007/s13369-019-03939-y
36. M. Awni, M. I. Khalil, and H. M. Abbas, "Deep-learning ensemble for offline arabic handwritten words recognition," in Proceedings - ICCES 2019: 2019 14th International Conference on Computer Engineering and Systems, Dec. 2019, pp. 40–45, doi: https://doi.org/10.1109/ICCES48960.2019.9068184.
37. H. Alyahya, M. M. Ben Ismail, and A. Al-Salman, "Deep ensemble neural networks for recognizing isolated Arabic handwritten characters," Accent. Trans. Image Process. Comput. Vis., vol. 6, no. 21, pp. 68–79, Nov. 2020, doi: https://doi.org/10.19101/tipcv.2020.618051.

38. H. M. Balaha, H. A. Ali, and M. Badawy, "Automatic recognition of handwritten Arabic characters: a comprehensive review," Neural Computing and Applications. Springer, pp. 1–24, Jul. 17, 2020, doi: https://doi.org/10.1007/s00521-020-05137-6.
39. M. Shams, A. A., and W. Z., "Arabic Handwritten Character Recognition based on Convolution Neural Networks and Support Vector Machine," Int. J. Adv. Comput. Sci. Appl., vol. 11, no. 8, pp. 144–149, 2020, doi: https://doi.org/10.14569/IJACSA.2020.0110819.
40. Al-Jourishi, A.A.; Omari, M.: Handwritten Arabic characters recognition using a hybrid two-stage classifier. Int. J. Adv. Comput. Sci. Appl. **11**(6), 143–148 (2020). https://doi.org/10.14569/IJACSA.2020.0110619
41. Ghanim, T.M.; Khalil, M.I.; Abbas, H.M.: Comparative Study on Deep Convolution Neural Networks DCNN-Based Offline Arabic Handwriting Recognition. IEEE Access **8**, 95465–95482 (2020). https://doi.org/10.1109/ACCESS.2020.2994290
42. Mustafa, M.E.; Elbashir, M.K.: A deep learning approach for handwritten Arabic names recognition. Int. J. Adv. Comput. Sci. Appl. **11**(1), 678–682 (2020). https://doi.org/10.14569/ijacsa.2020.0110183
43. R. Ahmad, S. Naz, M. Afzal, M. Liwicki, and A. Dengel, "A Deep Learning based Arabic Script Recognition System: Benchmark on KHAT," Int. Arab J. Inf. Technol., vol. 17, no. 3, 2020, doi: https://doi.org/10.34028/iajit/17/3/3.
44. Eltay, M.; Zidouri, A.; Ahmad, I.: Exploring Deep Learning Approaches to Recognize Handwritten Arabic Texts. IEEE Access **8**, 89882–89898 (2020). https://doi.org/10.1109/ACCESS.2020.2994248
45. I. Due Trier, A. K. Jain, and T. Taxt, "FEATURE EXTRACTION METHODS FOR CHARACTER RECOGNITION | A SURVEY," Pattern Recognit., vol. 29, no. 4, pp. 641–662, 1996.
46. Hicham, E.M.; Akram, H.; Khalid, S.: Using features of local densities, statistics and HMM toolkit (HTK) for offline Arabic handwriting text recognition. J. Electr. Syst. Inf. Technol. **4**(3), 387–396 (Dec. 2017). https://doi.org/10.1016/j.jesit.2016.07.005
47. R. Hussain, A. Raza, I. Siddiqi, K. Khurshid, and C. Djeddi, "A comprehensive survey of handwritten document benchmarks: structure, usage and evaluation," Eurasip Journal on Image and Video Processing, vol. 2015, no. 1. Springer International Publishing, pp. 1–24, Dec. 01, 2015, doi: https://doi.org/10.1186/s13640-015-0102-5.
48. M. Torki, M. E. Hussein, A. Elsallamy, M. Fayyaz, and S. Yaser, "WINDOW-BASED DESCRIPTORS FOR ARABIC HANDWRITTEN ALPHABET RECOGNITION: A COMPARATIVE STUDY ON A NOVEL DATASET."
49. A. El Sawy, H. El-Bakry, and M. Loey, "Arabic Handwritten Characters Dataset (AHCD)." .
50. "HACDB: Handwritten Arabic characters database for automatic character recognition - IEEE Conference Publication." https://ieeexplore-ieee-org.sdl.idm.oclc.org/document/6623974 (accessed Jan. 17, 2021).
51. H. M. Balaha, H. A. Ali, M. Saraya, and M. Badawy, "A new Arabic handwritten character recognition deep learning system (AHCR-DLS)," Neural Comput. Appl., pp. 1–43, Oct. 2020, doi: https://doi.org/10.1007/s00521-020-05397-2.
52. "(18) (PDF) IFN/ENIT-database of handwritten Arabic words." https://www.researchgate.net/publication/228904501_IFNENIT-database_of_handwritten_Arabic_words (accessed Jan. 13, 2021).
53. S. Al-Ma'adeed, D. Elliman, and C. Higgins, "A Data Base for Arabic Handwritten Text Recognition Research," 2004.
54. M. Musa, "Towards building competent dataset of Arabic recognition," Int. J. Eng. Adv. Res. Technol., vol. 2, no. 2, Feb. 2016, Accessed: Jan. 24, 2021. [Online]. Available: www.sustech.edu.
55. Mahmoud, S.A., et al.: KHATT: An open Arabic offline handwritten text database. Pattern Recognit. **47**(3), 1096–1112 (2014). https://doi.org/10.1016/j.patcog.2013.08.009
56. S. A. Mahmoud et al., "KHATT: An open Arabic offline handwritten text database," 2013, doi: https://doi.org/10.1016/j.patcog.2013.08.009.
57. K. Simonyan and A. Zisserman, "VERY DEEP CONVOLUTIONAL NETWORKS FOR LARGE-SCALE IMAGE RECOGNITION," 2015. Accessed: Nov. 15, 2020. [Online]. Available: http://www.robots.ox.ac.uk/.
58. K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Dec. 2016, vol. 2016-December, pp. 770–778, doi: https://doi.org/10.1109/CVPR.2016.90.
59. A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks." Accessed: Nov. 19, 2020. [Online]. Available: http://code.google.com/p/cuda-convnet/.
60. S. Liu and W. Deng, "Very deep convolutional neural network based image classification using small training sample size," in Proceedings - 3rd IAPR Asian Conference on Pattern Recognition, ACPR 2015, Jun. 2016, pp. 730–734, doi: https://doi.org/10.1109/ACPR.2015.7486599.
61. C. Szegedy et al., "Going deeper with convolutions," in Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Oct. 2015, vol. 07–12-June-2015, pp. 1–9, doi: https://doi.org/10.1109/CVPR.2015.7298594.
62. S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Nov. 2017, vol. 2017-January, pp. 5987–5995, doi: https://doi.org/10.1109/CVPR.2017.634.
63. Y. Zhu and S. Newsam, "DenseNet for dense flow," in Proceedings - International Conference on Image Processing, ICIP, Feb. 2018, vol. 2017-September, pp. 790–794, doi: https://doi.org/10.1109/ICIP.2017.8296389.
64. Cireşan, D.; Meier, U.; Masci, J.; Schmidhuber, J.: Multi-column deep neural network for traffic sign classification. Neural Netw. **32**, 333–338 (Aug. 2012). https://doi.org/10.1016/j.neunet.2012.02.023
65. N. Srivastava, G. Hinton, A. Krizhevsky, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," 2014. Accessed: Mar. 02, 2021. [Online]. Available: http://jmlr.org/papers/v15/srivastava14a.html.
66. A. Granet et al., "Trans-fer Learning for Handwriting Recognition on Historical Documents," Jan. 2018. Accessed: Jan. 27, 2021. [Online]. Available: https://hal.archives-ouvertes.fr/hal-01681126.
67. N. Srivastava, G. Hinton, A. Krizhevsky, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," 2014. Accessed: Apr. 04, 2021. [Online]. Available: http://jmlr.org/papers/v15/srivastava14a.html.
68. S. Ioffe and C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," PMLR, Jun. 2015. Accessed: Apr. 04, 2021. [Online]. Available: http://proceedings.mlr.press/v37/ioffe15.html.
69. I. J. Goodfellow, D. Warde-Farley, M. Mirza, A. Courville, and Y. Bengio, "Maxout Networks," in Proceedings of the 30th International Conference on Machine Learning, Atlanta, Georgia, USA, 2013, pp. 1319–1327.
70. A. Krogh· and J. A. Hertz, "A Simple Weight Decay Can Improve Generalization," in Proceedings of the 4th International Conference on Neural Information Processing Systems, 1991, pp. 950–957.
71. L. Wan, M. Zeiler, S. Zhang, Y. Lecun, and R. Fergus, "Regularization of Neural Networks using DropConnect," PMLR, May 2013.

Accessed: Apr. 04, 2021. [Online]. Available: http://proceedings.mlr.press/v28/wan13.html.
72. Alrobah N, Albahli S. A Hybrid Deep Model for Recognizing Arabic Handwritten Characters. IEEE Access. 2021 Jun 8.
73. Alsaffar, Ahmed & Awang, Suryanti & AL-Saiagh, Wafaa & Tiun, Sabrina & Al-Khaleefa, Ahmed Salih. (2018). Deep Learning Algorithms for Arabic Handwriting Recognition: A Review. International Journal of Engineering & Technology. 7. 344. https://doi.org/10.14419/ijet.v7i3.20.19271.
74. Musa, Mohamed Elhafiz. (2011). Arabic handwritten datasets for pattern recognition and machine learning. 1–3. https://doi.org/10.1109/ICAICT.2011.6110959.
75. Noubigh, Zouhaira & Anis, Mezghani. (2021). Contribution on Arabic Handwriting Recognition Using Deep Neural Network. https://doi.org/10.1007/978-3-030-49336-3_13
76. Alkhawaldeh, R.: Arabic (Indian) digit handwritten recognition using recurrent transfer deep architecture. Soft. Comput. **1–11**,(2021). https://doi.org/10.1007/s00500-020-05368-8